Optimization and Uncertainty in Learning Report

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Neural Network. Optimization and Stability

***Abstract*—This study isolates how optimization and regularization choices affect training dynamics, seed sensitivity, runtime, and class-balanced generalization for a fixed PyTorch multilayer perceptron on the seven-class Covertype task. The data split, preprocessing, class-weighted loss, architecture, batch size, and hardware class were held constant. Randomized hill climbing (RHC), simulated annealing, and a genetic algorithm optimized only the 455-parameter output layer; seven gradient optimizers were compared under 1,600-update budgets; and L2 weight decay, dropout, early stopping, label smoothing, and input noise were evaluated with standard Adam. Results show that lower validation loss did not reliably imply higher Macro-F1. RHC reduced median validation loss after SGD but decreased median test Macro-F1. Adam without bias correction obtained the highest median test Macro-F1 (0.5824) and crossed the loss threshold fastest, but it was substantially more seed-sensitive than standard Adam. L2 produced a small Macro-F1 gain while dropout combinations reduced the train–validation gap yet harmed multiclass F1. A final RHC pass added evaluations without a reliable paired gain. The practical conclusion is to select training recipes with seed-aggregated Macro-F1, balanced accuracy, per-class behavior, and compute cost rather than accuracy or validation loss alone.**

***Index Terms*—Adam, randomized optimization, regularization, Macro-F1, class imbalance, Covertype, stability**

I. Introduction and Hypothesis

Optimizer comparisons are easily confounded by changes in architecture, preprocessing, training duration, or hyperparameter-search effort. This paper instead asks a controlled question: given the same data representation and neural backbone, which training choices change convergence speed, stability, and multiclass generalization, and which merely improve a surrogate objective?

Covertype is strongly imbalanced: the held-out set contains 273,549 examples of class 2 but only 2,653 of class 4. Accuracy therefore rewards majority-class decisions more heavily than rare-class behavior. Macro-F1 gives each class equal weight after computing its class-specific F1, while balanced accuracy averages class recalls. Accuracy is retained, but it is not the primary selection criterion.

Three hypotheses were specified before the full runs. First, search over the final layer might slightly improve validation loss, but would offer weak test Macro-F1 return per function evaluation. Second, Adam-family methods would reach a fixed validation-loss threshold faster than plain SGD; momentum would reduce SGD variance; and removing Adam bias correction was expected to hurt early training. Third, moderate regularization would reduce overfitting and affect rare-class performance more than raw accuracy. The final experiment composed the strongest validation-selected elements and tested whether their interaction survived three seeds.

TABLE I
Controlled protocol. *Takeaway:* The design changes optimizer or regularizer behavior while holding the representation, split, backbone, loss, and principal budgets fixed.

Item	Fixed setting
Data	16,000 train; 4,000 validation; 561,012 locked test
Preprocessing	Train-fitted standardization; stable 1–7 label map
Backbone	54–64–64–7 ReLU MLP; 8,135 parameters
Loss / batch	Class-weighted cross-entropy / 512
Seeds / hardware	42, 202, 7641 / CPU, 4 PyTorch threads
RO budget	Final layer only (455 parameters); 300 objective calls; fixed 2,000-row stratified validation subset
Gradient budget	1,600 updates for Parts 2–4; same evaluation cadence

II. Data and Experimental Design

A. Data, Model, and Metrics

The 20,000-row stratified development sample was divided into 16,000 training and 4,000 validation records. The remaining 561,012 records formed the locked test set. An exact row-multiset audit verified that the development sample plus the remainder reconstructed all 581,012 Covertype rows. The 54-feature standardization transform was fitted only on training data and then applied to validation and test data.

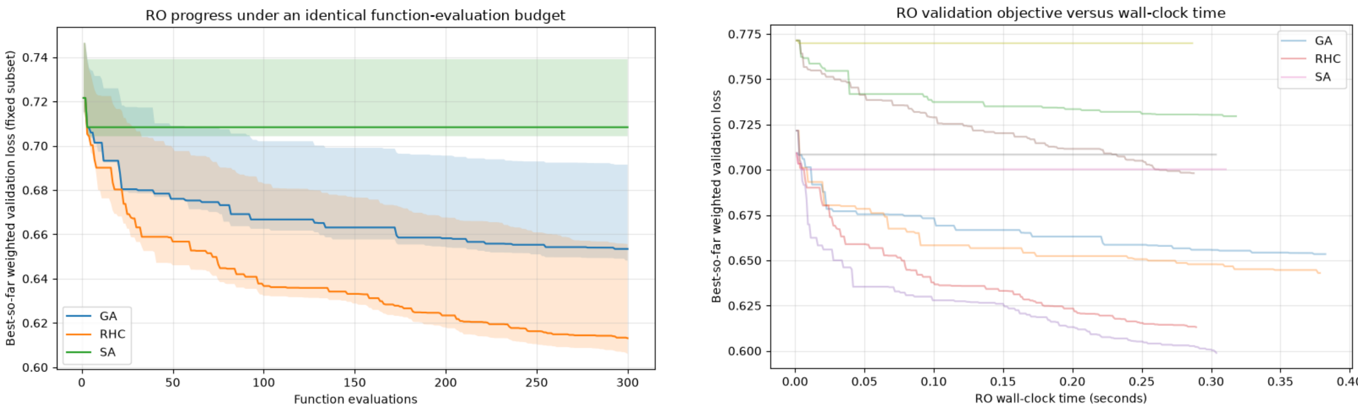
The fixed network was $54 \to 64 \to 64 \to 7$, with ReLU after each hidden layer and 8,135 parameters. Training used a batch size of 512 and class-weighted cross-entropy. The reported outcomes are validation loss, accuracy, Macro-F1, balanced accuracy, per-class precision/recall/F1, gradient evaluations, function evaluations, and wall time. Runs used deterministic PyTorch settings, four CPU threads, and seeds 42, 202, and 7641.

III. Methods

A. Optimizer and Regularization Protocols

Part 1 pretrained the backbone using the supervised-learning SGD configuration and early stopping on validation Macro-F1. RHC, simulated annealing (SA), and a genetic algorithm (GA) then optimized the output layer. Each validation-objective call counted as one function evaluation; algorithm selection used the full validation set after the fixed-subset search.

Part 2 compared plain SGD, SGD with momentum, Nesterov, Adam, Adam without bias correction, Adam with $\beta_1 = 0$, and AdamW. To isolate optimizer ingredients from explicit regularization, all non-AdamW conditions used zero weight decay; the weight-decay study was reserved for Part 3, consistent with the experimental-control interpretation discussed in [**?**]. All seven conditions received 1,600 updates. Learning rates were



(a) Best-so-far objective versus function evaluations.

(b) Objective versus RO wall time.

Fig. 1. Randomized optimization under identical 300-call budgets. *Takeaway:* RHC is the most compute-efficient validation-loss search, but the result must be judged against Macro-F1 because objective descent alone did not generalize.

selected on validation data, and a threshold of $\ell = 0.599776$ was locked before the multi-seed comparison. Coarse 3×3 Adam sensitivity grids varied (α, β_1) and (α, β_2).

Part 3 fixed standard Adam at $\alpha = 0.003$, $\beta_1 = 0.9$, and $\beta_2 = 0.99$. Individual sweeps tested weight decay, hidden-layer dropout, label smoothing, training-only Gaussian input noise, and early stopping. Part 4 used weight decay 10^{-4}, tested three small dropout interactions (0.05, 0.10, 0.15), locked 0.15, and applied the selected RHC method for only 30 function evaluations.

IV. Randomized Optimization

RHC produced the lowest median full-validation loss, 0.6505, compared with 0.6555 for GA and 0.7102 for SA. Its median RO-only time was also lowest at 0.289 s. Figure 1 shows that most improvement arrived early, after which progress became incremental. GA improved quickly but retained a higher objective; SA accepted many candidates but did not move to the same low-loss region.

The objective improvement did not translate to the primary test metric. Median test Macro-F1 fell from 0.5424 for SGD pretraining to 0.5219 after RHC, 0.5329 after SA, and 0.5210 after GA. RHC did raise median balanced accuracy from 0.6931 to 0.7190. This divergence is explained by class-level behavior: the RHC representative run achieved high recalls for rare classes (0.846 for class 4, 0.823 for class 5, and 0.888 for class 7) but low precisions (0.287, 0.139, and 0.410). More minority instances were recovered, yet many majority examples were reassigned to rare classes, lowering class F1 and overall accuracy. Thus, RO found a lower weighted validation-loss region that was not a better multiclass decision rule.

V. Adam-Family Ablations

A. Sensitivity and Convergence Speed

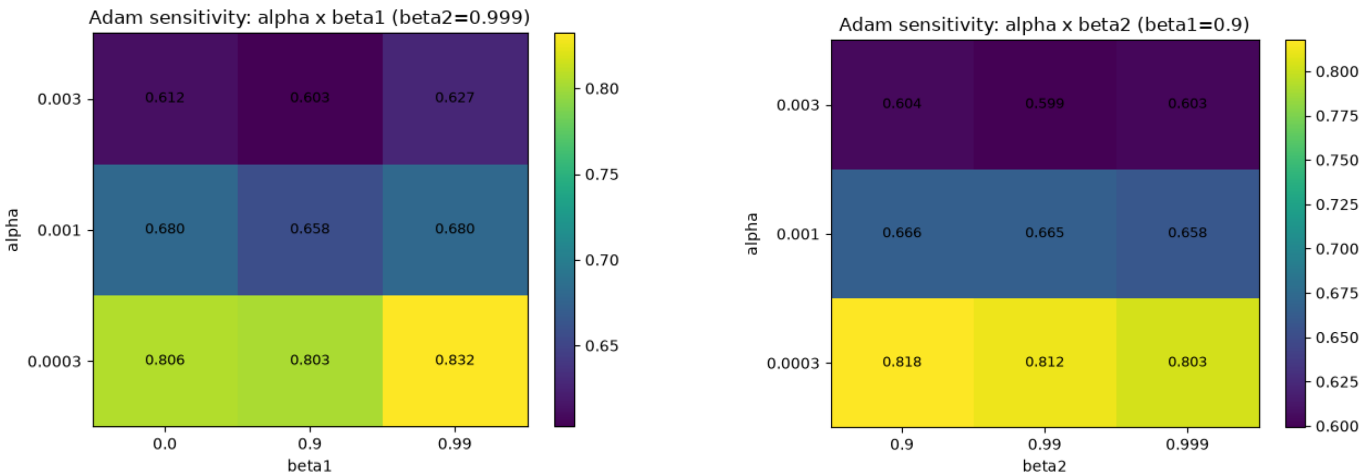
The heatmaps in Fig. 2 show that learning rate dominated the coarse Adam grid. At $\alpha = 0.003$, $\beta_1 = 0.9$ and $\beta_2 = 0.99$ gave the strongest standard-Adam validation loss. Lower learning rates underused the fixed budget, while $\beta_1 = 0.99$ slowed adaptation enough to degrade the objective.

Plain SGD never reached ℓ. Standard Adam reached it in a median 608 updates and 1.669 s; momentum SGD required 832 updates and 1.895 s. Adam without bias correction reached it in 320 updates and 0.824 s. The trajectory plots in Fig. 3 confirm that adaptive methods rapidly entered the useful loss range, whereas plain SGD remained higher and noisier. Setting $\beta_1 = 0$ retained adaptive second-moment scaling but crossed the threshold only after 1,088 median updates, indicating that first-moment accumulation mattered for early progress.

B. Generalization and Stability

Adam without bias correction achieved the best median test Macro-F1, 0.5824, and a median accuracy of 0.6765. It also reached the fixed validation-loss threshold faster than expected, an outcome independently raised in the course discussion in [**?**]. However, its test Macro-F1 standard deviation was 0.0224, compared with 0.0010 for standard Adam. The uncorrected update changed the early step schedule and happened to fit two seeds well, but its wide 0.5492–0.5918 range marks it as a validation-dependent, higher-risk choice rather than a universal improvement. This directly rejects the prediction that removing bias correction would necessarily be harmful; the result is instead a speed–stability trade-off.

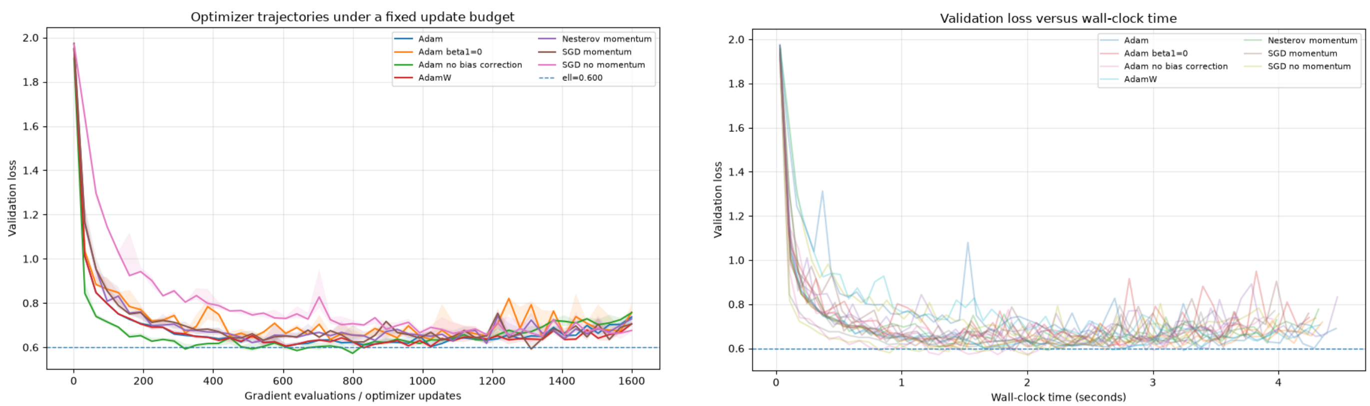
Momentum SGD nearly matched standard Adam in median test Macro-F1 (0.5666 versus 0.5659) with lower runtime, and its F1 variability was far below plain SGD. AdamW obtained the highest median accuracy (0.6827) and balanced accuracy (0.7533), but only 0.5661 Macro-F1. Decoupled shrinkage therefore improved aggregate classification without consistently improving equal-weight class F1. Nesterov was less reliable than classical momentum on this network, indicating that acceleration was not automatically useful at the chosen learning rate and budget.



(a) $\alpha \times \beta_1$ with $\beta_2 = 0.999$.

(b) $\alpha \times \beta_2$ with $\beta_1 = 0.9$.

Fig. 2. Coarse Adam validation-loss sensitivity. *Takeaway:* The fixed budget required $\alpha = 0.003$; moderate first-moment memory and $\beta_2 = 0.99$ were preferable to more inertial settings.



(a) Validation loss versus optimizer updates.

(b) Validation loss versus wall-clock time.

Fig. 3. Seven optimizer trajectories across three seeds. *Takeaway:* Adaptive scaling and momentum reduce time to a useful loss region, but the fastest condition is not the most stable one.

VI. Regularization

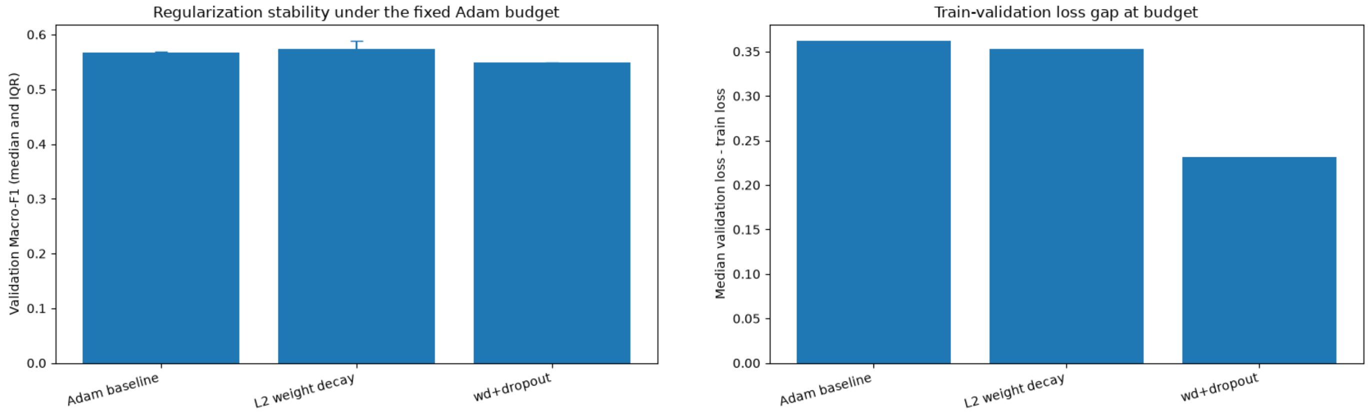
The regularization sweep produced a key distinction between fitting stability and predictive utility. Early stopping reduced median updates to 640 and the train–validation loss gap to 0.134, but median validation Macro-F1 fell to 0.5573. Dropout reduced the gap from 0.362 to 0.242, yet its median validation Macro-F1 was 0.5553. Label smoothing was especially harmful: median validation loss rose to 1.484 and Macro-F1 fell to 0.496. With class-weighted cross-entropy, smoothing diluted the hard class targets that were compensating for imbalance, creating underconfident decisions without a compensating generalization gain.

L2 weight decay was the best individual regularizer. Relative to the Adam baseline, it improved median test Macro-F1 from 0.5659 to 0.5697 and accuracy from 0.6564 to 0.6599, while balanced accuracy changed slightly from 0.7464 to 0.7453. The gain was not stable: test Macro-F1 standard deviation increased from 0.0010 to 0.0128. The weight-decay plus dropout combination reduced the validation loss and generalization gap, but test Macro-F1 dropped to 0.5480. Figure 4 shows why selecting only by gap reduction would have been misleading.

Class-level results clarify the aggregate metrics. Weight decay improved class-4 F1 from 0.5200 to 0.5541 and class-7 F1 from 0.6303 to 0.6591, but class-2 F1 fell from 0.6892 to 0.6495. The dropout combination increased rare-class recall but reduced precision, recreating the same failure mode observed in RO. Regularization therefore redistributed errors rather than uniformly improving the classifier.

VII. Integrated Comparison

Table II summarizes median results across seeds. The highest Macro-F1 method was the no-bias-correction Adam ablation, not the integrated recipe. Standard Adam was the most stable optimizer; AdamW was preferable when accuracy or balanced accuracy had priority. L2 was the only regularizer with a



(a) Median validation Macro-F1 with seed uncertainty.

(b) Median train–validation loss gap.

Fig. 4. Regularization effects at the fixed Adam budget. *Takeaway:* Weight decay gives a small class-balanced gain, while dropout chiefly reduces overfitting indicators; combining them lowers the gap but harms test Macro-F1.

TABLE II
MEDIAN VALIDATION, TEST, AND COMPUTE OUTCOMES ACROSS THREE SEEDS. GE DENOTES GRADIENT EVALUATIONS AND FE DENOTES VALIDATION-OBJECTIVE FUNCTION EVALUATIONS. *Takeaway:* THE BEST VALIDATION-LOSS, ACCURACY, MACRO-F1, BALANCED-ACCURACY, AND RUNTIME CHOICES ARE DIFFERENT; NO SINGLE SCALAR CAPTURES THE OPTIMIZER TRADE-OFF.

Method	Val. loss	Accuracy	Macro-F1	Bal. acc.	Time (s)	GE	FE
SL SGD pretraining	0.7203	0.6401	0.5424	0.6931	1.620	1152	0
SGD + RHC	0.6505	0.6243	0.5219	0.7190	1.909	1152	300
SGD momentum	0.5884	0.6642	0.5666	0.7416	3.651	1600	0
Standard Adam	0.5990	0.6564	0.5659	0.7464	4.320	1600	0
Adam, no bias correction	**0.5712**	0.6765	**0.5824**	0.7501	3.970	1600	0
Adam, $\beta_1 = 0$	0.6028	0.6657	0.5640	0.7477	4.244	1600	0
AdamW	0.5927	**0.6827**	0.5661	**0.7533**	4.349	1600	0
Adam + L2	0.5935	0.6599	0.5697	0.7453	4.342	1600	0
Adam + L2 + dropout	0.5833	0.6411	0.5480	0.7514	4.866	1600	0
Integrated before RO	0.5780	0.6534	0.5626	0.7512	5.319	1600	0
Integrated + RHC	0.5784	0.6468	0.5638	0.7489	5.349	1600	30

TABLE III
REPRESENTATIVE PER-CLASS F1 ON THE LOCKED TEST SET. THE STANDARD AND NO-BIAS ADAM REPORTS USE THE MEDIAN-MACRO-F1 SEED; L2 AND INTEGRATED COLUMNS USE THEIR REPRESENTATIVE MEDIAN SEEDS. *Takeaway:* AGGREGATE GAINS COME FROM DIFFERENT CLASS REDISTRIBUTIONS: NO-BIAS ADAM IMPROVES CLASSES 1, 2, 5, 6, AND 7, WHEREAS L2 HELPS CLASSES 4 AND 7 BUT SACRIFICES THE DOMINANT CLASS 2.

Class (support)	Adam	Adam no-bias	Adam + L2	Integrated + RHC
1 (204,548)	0.7028	**0.7263**	0.7163	0.7022
2 (273,549)	0.6892	**0.6965**	0.6495	0.6579
3 (34,523)	**0.6722**	0.6574	0.6570	0.6634
4 (2,653)	0.5200	0.5158	**0.5541**	0.5430
5 (9,166)	0.2503	**0.2812**	0.2466	0.2768
6 (16,769)	0.4911	**0.5148**	0.5050	0.4914
7 (19,804)	0.6303	**0.6845**	0.6591	0.6117

positive median Macro-F1 movement, and the improvement was small.

The final 30-call RHC pass changed the paired median test Macro-F1 by -0.0005 and balanced accuracy by -0.0023. Figure 5 shows almost flat best-so-far trajectories for two seeds and only a small reduction for one. The function-evaluation cost was small in seconds but produced no robust predictive benefit. The integrated recipe therefore failed to exceed the best Part 2 result.

The representative confusion matrices reinforced the same pattern: most error mass remained on the class-1/class-2 boundary, while rare-class recall was high but precision remained limited. Consequently, balanced accuracy could remain high even when Macro-F1 and raw accuracy fell.

VIII. DISCUSSION AND HYPOTHESIS RESOLUTION

The RO hypothesis was supported only for the surrogate objective. RHC efficiently lowered validation loss, but its test Macro-F1 return was negative. Search on a small final layer can therefore exploit the weighted loss without improving the full decision boundary. The appropriate rule is to retain RO only when gains survive full-validation evaluation, seed aggregation, and per-class analysis.

The optimizer-speed hypothesis was accepted. Adam crossed the validation threshold while plain SGD did not, and mo-

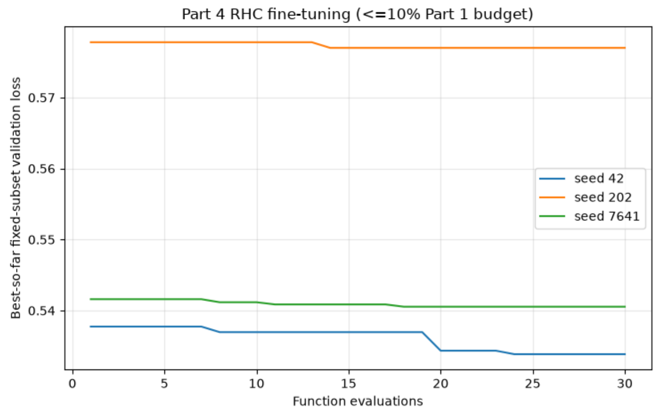


Fig. 5. RHC fine-tuning with 30 calls. *Takeaway:* The final-layer search is essentially flat at the reduced budget and does not justify inclusion in the deployed recipe.

mentum reduced the large variance of plain SGD. The bias-correction sub-hypothesis was rejected: the no-correction variant was fastest and had the best median Macro-F1, although its seed variance was high. Bias correction remains the more reliable default because the apparent advantage was not uniformly reproduced.

The regularization hypothesis was qualified. L2 produced a small Macro-F1 gain and improved several minority-class F1 scores, but it increased seed sensitivity and slightly reduced balanced accuracy. Early stopping and dropout clearly reduced train–validation gaps, yet those improvements did not imply better test Macro-F1. The combined recipe over-regularized the fixed compact network and shifted probability mass toward minority labels, improving recall-oriented measures while reducing precision.

The resulting decision rule is: use standard Adam at $\alpha = 0.003$, $\beta_1 = 0.9$, and $\beta_2 = 0.99$ when stability is the priority; consider the no-bias variant only after repeated-seed validation confirms its advantage; use AdamW when accuracy and balanced accuracy dominate the objective; add 10^{-4} weight decay only when its minority-class gains survive seed aggregation; and reject final-layer RO or dropout combinations when their validation-loss gains are not accompanied by Macro-F1 improvement.

IX. Conclusion

Controlled budgets revealed that optimization ingredients affected convergence more strongly than final generalization, while regularization often changed error allocation rather than overall correctness. The strongest median test Macro-F1 came from Adam without bias correction, but standard Adam was markedly more stable. RHC and dropout combinations improved selected validation or gap statistics without improving the primary test metric. The central lesson is that an imbalanced multiclass neural classifier should be selected using seed-aggregated Macro-F1, balanced accuracy, per-class precision/recall, and compute cost together. A lower validation loss or smaller generalization gap is evidence about training behavior, not proof of a better classifier.